

| Command | Sample Usage            | What It Does                                                                                       |
|---------|-------------------------|----------------------------------------------------------------------------------------------------|
| pwd     | pwd                     | Prints the full path of the current directory (Print Working Directory)                            |
| ls      | ls -la                  | Lists files and folders in the current directory (-la shows hidden files with details)             |
| cd      | cd Documents            | Changes the current directory to the one specified                                                 |
| mkdir   | mkdir projects          | Creates a new folder named "projects"                                                              |
| touch   | touch notes.txt         | Creates a new empty file named "notes.txt"                                                         |
| cp      | cp file1.txt backup.txt | Copies a file (creates "backup.txt" as a copy of "file1.txt")                                      |
| mv      | mv notes.txt archive/   | Moves a file into another folder, or renames it if the destination is a filename                   |
| rm      | rm oldfile.txt          | Deletes a file permanently (use <code>rm -r foldername</code> to delete a folder and its contents) |
| cat     | cat notes.txt           | Displays the entire contents of a file in the terminal                                             |
| grep    | grep "error" log.txt    | Searches for a specific word or pattern inside a file (here, finds lines containing "error")       |
| chmod   | chmod +x script.sh      | Changes a file's permissions (here, makes "script.sh" executable)                                  |
| whoami  | whoami                  | Changes a file's permissions (here, makes "script.sh" executable)                                  |

|            |                           |
|------------|---------------------------|
| git init   | git restore               |
| git status | git branch                |
| git add    | git checkout / git switch |
| git commit | git merge                 |
| git log    | git tag                   |
| git diff   | git clone                 |

Git manages your files across three "trees" (areas):

### 1. Working Directory

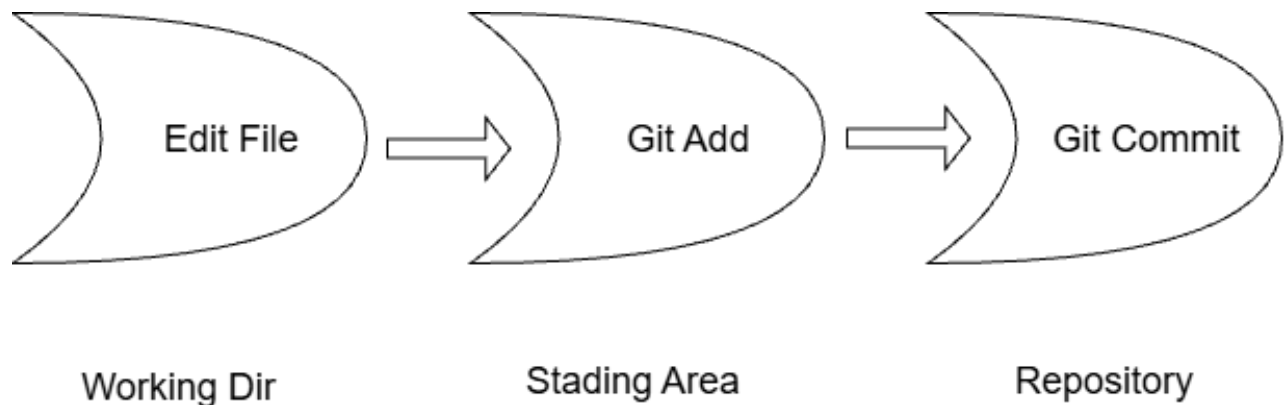
This is your actual project folder — the files you can see and edit on your computer right now. Any change you make here (creating, editing, deleting files) only exists locally and isn't tracked by Git yet.

### 2. Staging Area (Index)

A holding area where you place changes you want to include in your next commit. Running `git add <file>` moves a change from the Working Directory into the Staging Area. Think of it as a "preview" of your next snapshot — you can add or remove files from it before committing.

### 3. Repository (`.git` directory)

The permanent history of your project. Running `git commit` takes everything in the Staging Area and saves it as a new snapshot inside the `.git` folder. This is what makes the change official and trackable — you can view it later with `git log`.



## References

Atlassian. (n.d.). Git commit. Atlassian Git Tutorial. Retrieved September 18, 2026, from <https://www.atlassian.com/git/tutorials/saving-changes/git-commit>

Chacon, S., & Straub, B. (2014). Pro Git (2nd ed.). Apress.  
<https://git-scm.com/book/en/v2>

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Tools used: AI Claude for Research of sources